

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau
Practical – X

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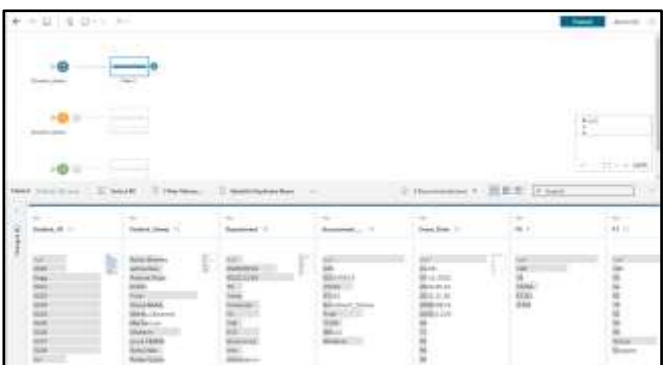
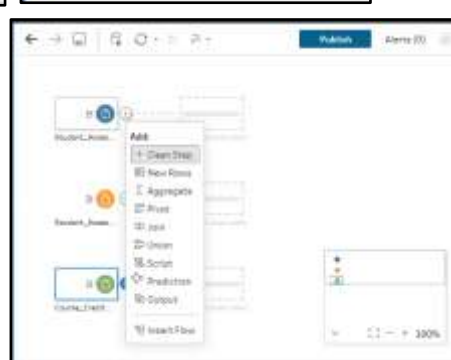
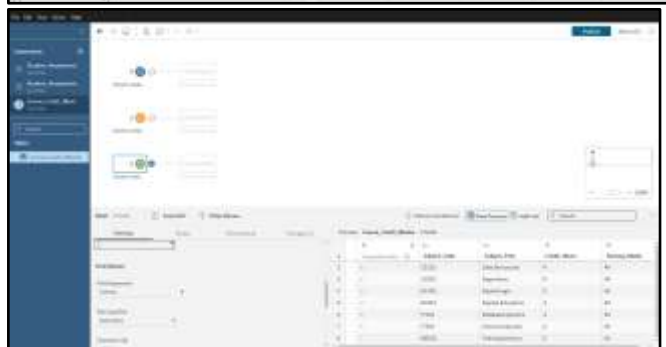
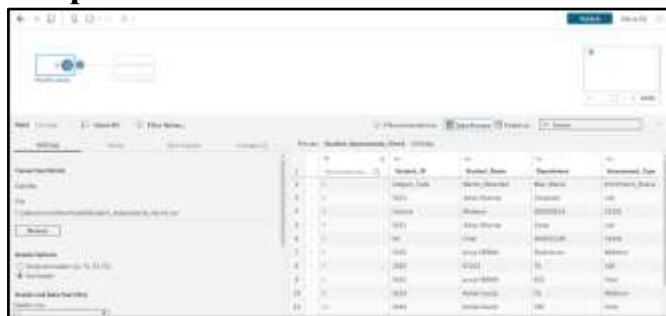
Data Preparation and Transformation using Tableau Prep

A) Connect and load data into Tableau Prep

Steps:-

1. Open Tableau Prep Builder.
2. Click Connect to Data.
3. Select Text File and open the required **Student_Assessment** file.
4. In Text Options, select Comma as the field separator.
5. Add the required table to the Flow Pane to create the Input Step.
6. Check the data preview and verify fields such as Student_ID, Student_Name, Department, Assessment_Date, Score, and Max_Marks.
7. Keep the data ready for the next cleaning step.

Output:-



B) Clean and profile data

Steps:

1. Select the Input Step.
2. Click the + icon and select Clean Step.
3. Review the data profile and check the values of all fields.
4. Identify null, incorrect, duplicate, or inconsistent values.
5. Open the field menu and check/change the Data Type where required.
6. Change Assessment_Date to the Date data type.
7. Review the cleaned data and verify that the fields are in the correct format.

Output:-

The screenshots illustrate the steps taken in Alteryx Designer to clean and profile data. The first screenshot shows the 'Clean' step configuration menu. The second screenshot shows the 'Data Profile' view with a table of data. The third and fourth screenshots show the 'Data Type' dropdown menu for the 'Assessment' and 'Exam Date' fields, respectively, with 'Date & Time' and 'Date' selected. The fifth and sixth screenshots show the 'Data Type' dropdown menu for the 'Assessment' and 'Exam Date' fields, respectively, with 'Date' selected.

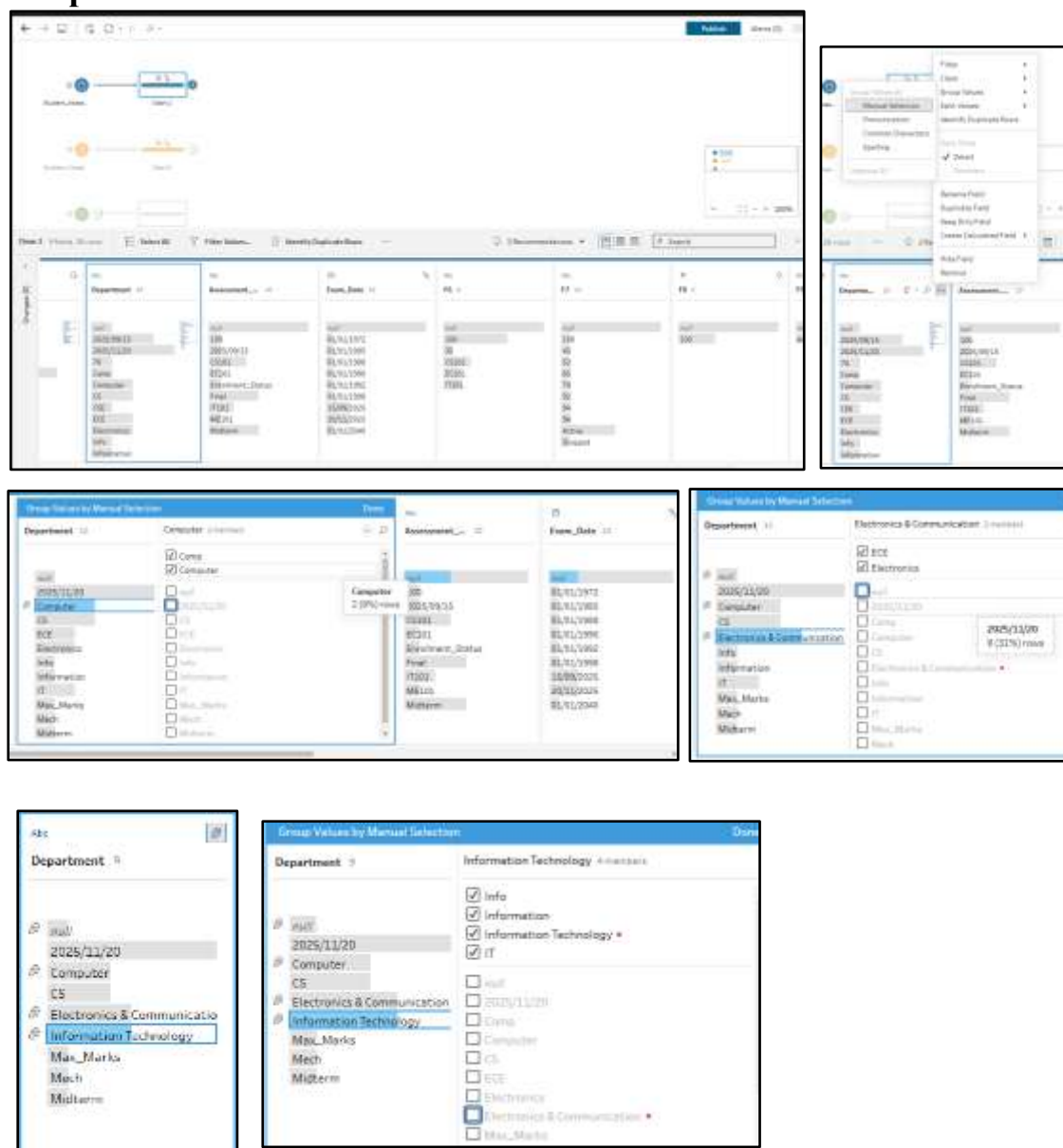
C) Group and standardize values

Steps:

1. Open the Clean Step.
2. Select the Department field.
3. Click the field's three-dot menu.
4. Select Group Values → Manual Selection.

5. Select similar department values, such as Computer/Computer Science, Electronics & Communication, and Information Technology/IT.
6. Combine the selected values into one standardized group.
7. Click Done and verify the standardized Department values.

Output:-

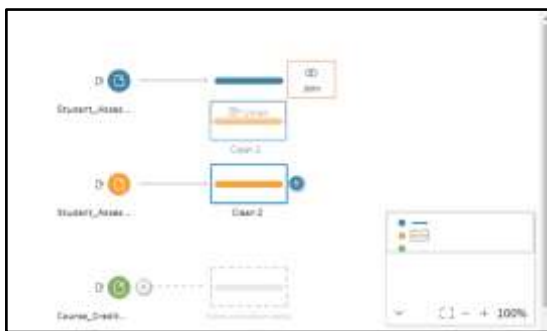


D) Merge mismatched fields

Steps:

1. Add the required Student_Assessment datasets to the Tableau Prep flow.
2. Create a Clean Step for each dataset.
3. Connect the Clean Steps of the two datasets.
4. Create Union 1 to combine the datasets.
5. Check the fields in the Union result, such as Table Names, F8, F9, Student_ID, and Student_Name.
6. Check whether the fields from both datasets are properly combined.
7. Verify the final Union result before moving to the next step.

Output:-




E) Perform aggregations in Prep

Steps:

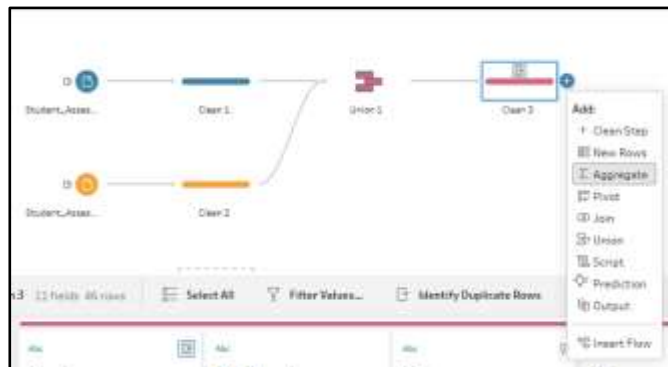
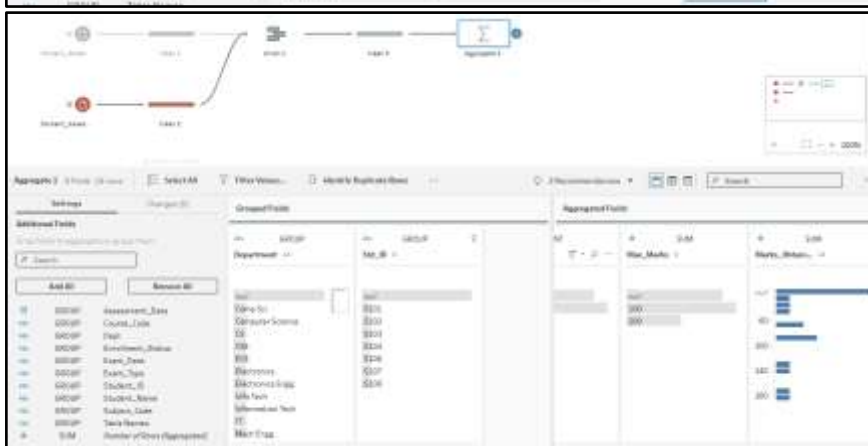
1. Select Clean 3 after the Union Step.
2. Click the + icon.
3. Select Aggregate.
4. Drag Department and Std_ID into Grouped Fields.
5. Drag Score into Aggregated Fields.
6. Set **Score** to Average (AVG).
7. Set **Full_Name** to Count (CNT) and **Max_Marks** to Sum (SUM) as shown in the screenshots.
8. Check the aggregated results in the profile pane.

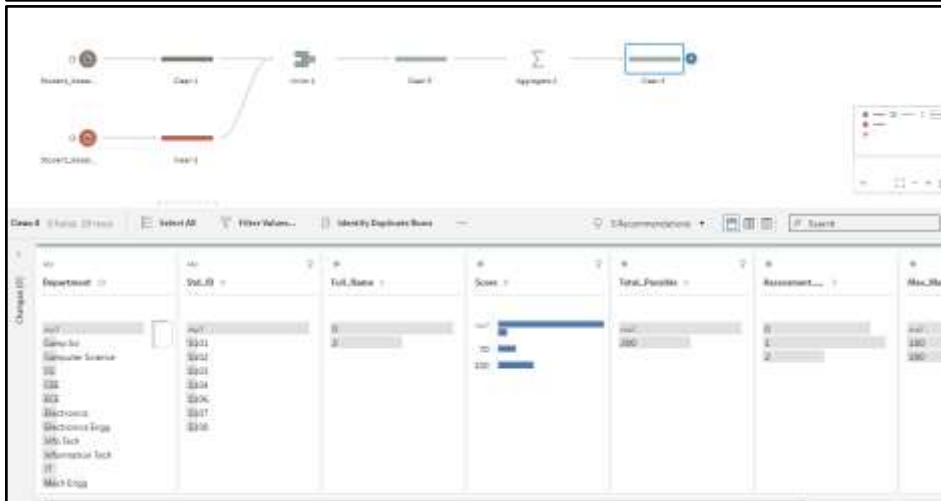
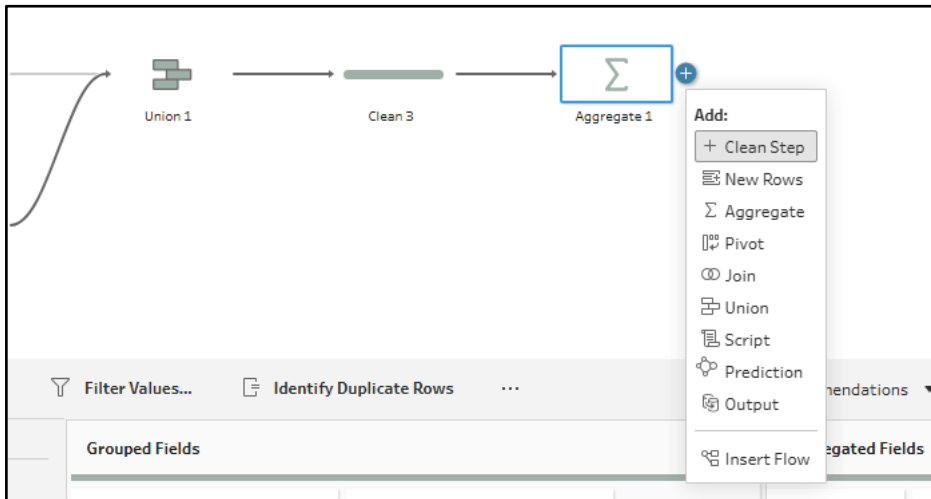
Output:-

```

IF (@@VERSION < 100000)
  PRINT 'This version of SQL Server is not supported.'
ELSE
  PRINT 'This version of SQL Server is supported.'
END

```

[illegible]



F) Apply filters and transformations.

Filter

Steps:

1. Select Clean 4.
2. Select the Student_Name field.
3. Open the field menu.
4. Select Filter → Selected Values.
5. Select the required value; in the screenshot, 2 is selected.
6. Click Done.
7. Verify the filtered result.

Formula 1 – Term Percentage

Steps:

1. Open the field menu and select Create Calculated Field.
2. Enter the Field Name as Term_percentage.
3. Enter the following formula: $\text{ROUND}([\text{Score}] / [\text{Total_Possible}] * 100, 2)$
4. Check that the calculation is valid.
5. Click Apply and Close.
6. Verify the newly created Term_percentage field.

Formula 2 – Letter Grade

Steps:

1. Select Create Calculated Field.
2. Enter the Field Name as Letter_Grade.

3. Enter the formula:

```

IF [Term_percentage] >= 85 THEN "A (Distinction)"
ELSEIF [Term_percentage] >= 75 THEN "B (First Class)"
ELSEIF [Term_percentage] >= 60 THEN "C (Second Class)"
ELSEIF [Term_percentage] >= 40 THEN "D (Pass Class)"
ELSE "F (Fail / Re-appear)"
END

```

4. Check that the calculation is valid.
5. Click Apply and Close.
6. Verify the **Letter_Grade** field.

Formula 3 – Academic Standing**Steps:**

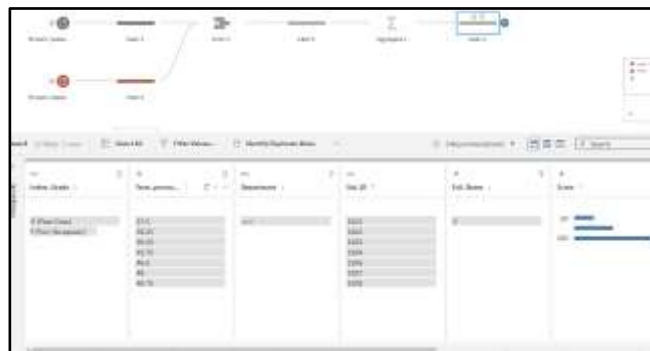
1. Select Create Calculated Field.
2. Enter the Field Name as Academic_Standing.
3. **Enter the formula:**

```

IF [Term_percentage] >= 50 THEN "Satisfactory"
ELSE "Academic Probation (<60%)"
END

```
4. Check that the calculation is valid.
5. Click Apply and Close.
6. Verify the **Academic_Standing** field.

Output:-

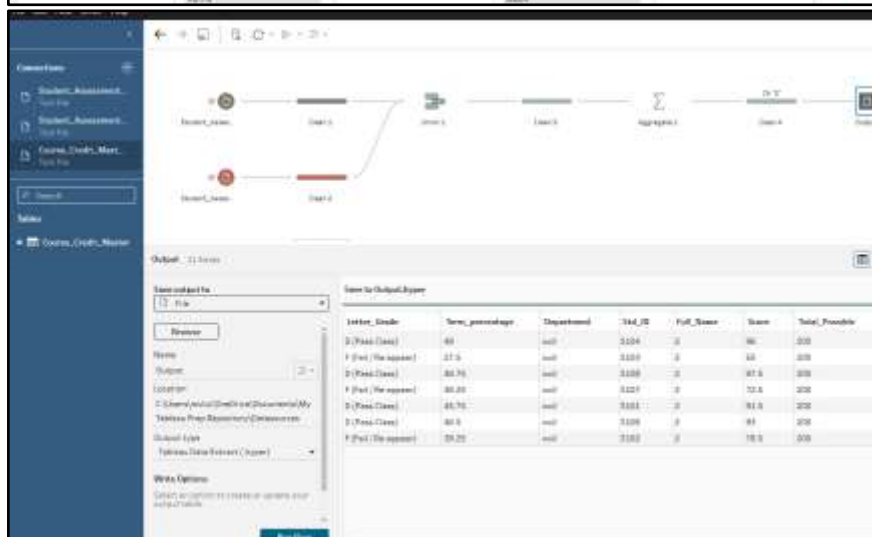


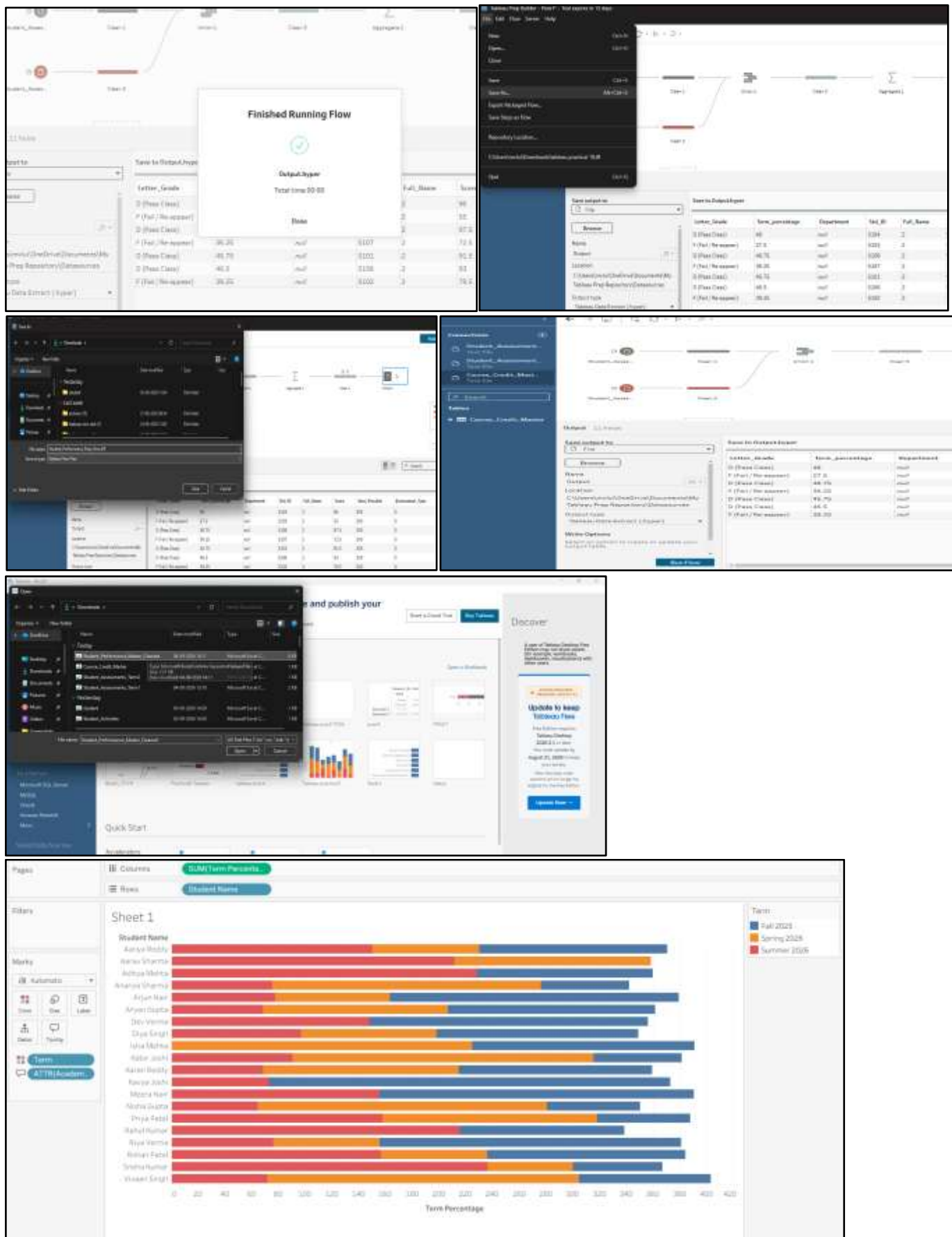
G) Create and automate data preparation flows.

Steps:

1. Review the complete flow: Input → Clean → Union → Clean → Aggregate → Clean.
2. Select the final Clean 4 step.
3. Click the + icon on the right side of Clean 4.
4. Select Output from the menu.
5. Set Save Output Type to File.
6. Enter the required Output Name and Location.
7. Check the output preview containing fields such as Letter_Grade, Term_Percentage, Department, Std_ID, Full_Name, Score, and Total_Possible.
8. Click Run Flow to execute the complete data preparation flow.
9. Verify the generated output dataset.

Output:-





Overall Conclusion

The complete data preparation process was performed successfully in Tableau Prep. The data was loaded, cleaned, standardized, combined, aggregated, filtered, and transformed using calculated fields. Finally, the prepared data was sent to an Output Step and the complete flow was executed successfully.